8TH IEEE SEASONAL SCHOOL ON

DPVSA

DIGITAL PROCESSING OF VISUAL SIGNALS AND APPLICATIONS

StaticBAC V2: Optimized binary arithmetic coding and entropy-aware quantization for neural network compression

Jiovana Gomes[1], Fábio Ramos[2], Sergio Bampi[1]

[1] UFRGS, Brazil
[2] Unipampa, Brazil

jsgomes@inf.ufrgs.br

1. INTRODUCTION

- Modern neural networks (NNs) have **millions to trillions of parameters** → Model **storage and memory transfers dominate cost** → Memory access is more expensive than computation
- Increasing deployment of NNs on embedded, edge devices → **strict memory capacity, bandwidth, latency requirements**
- **Need for efficient NN compression** → reduce model size and parameter traffic → enable execution on resource-constrained hardware

2. CONTEXT

- **StaticBAC V1 [1]** → simplified **binary arithmetic coding** (BAC) based on DeepCABAC for neural network model compression
- **Deep CABAC [2]** (Context Adaptive BAC) → **adaptive probability modeling** to achieve excellent compression but leads to **complex decoding** flow with limited throughput
- StaticBAC → **single fixed probability model** for simpler and faster decoding, trading **compression loss for speed and lower memory usage**
- **NEW: StaticBAC V2** → **Improves compression by adding more complexity, still static, a bit slower**

3. PROPOSAL

- StaticBAC V1 old coding flow: processes NN tensors in **chunks.**
  - Chunk mean is <u>always</u> subtracted from parameters (**residual coding**).
  - Simple and pessimist Bits per Element (BPE) cost to decide SKIP chunk.
  - **Binarization:** Must convert parameters into binary flags (bins) before BAC
- **NEW: StaticBAC V2** reworked binarization and remainder coding → groups of GT flags with separate probabilities. SIGN now bypass. See Fig. 1
- Improved chunk coding:
  - added <u>Neighbor</u> subtractor besides <u>Mean</u> and <u>None</u>
  - **More accurate BPE** compares subtractors and uses the probability model to estimate entropy cost (pre-computed cost table, not online)
- Updated the fixed probability table with StaticBAC statistics.
  - V1 was based on DeepCABAC statistics.

4. METHODS

- We compare StaticBAC V1 vs V2 using **3 separate quantization methods.**
  - **OLD** V1 mixed-width uniform MSE quantizer (INT8 for weights, INT12/16 all else)
  - INT8 all with uniform **MSE** quantizer (least reconstruction error)
  - INT8 all with uniform **rate-distortion (RD)** quantizer, λ=0.15 (reduces entropy)
- 13 neural networks evaluated. Original models in FP32.

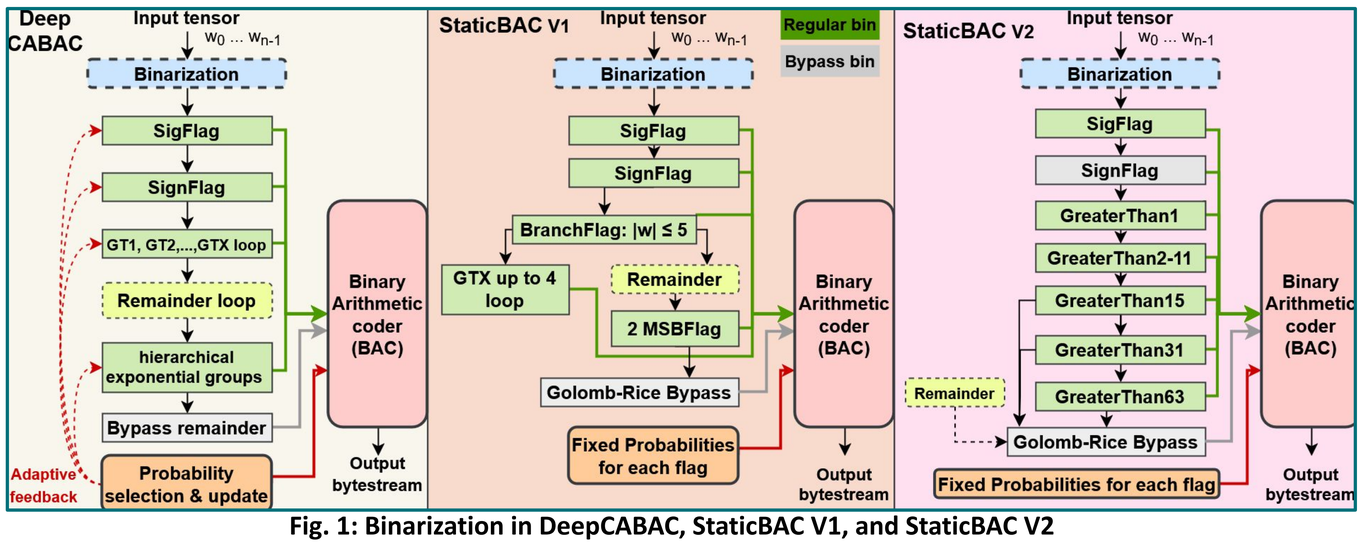


Fig. 1: Binarization in DeepCABAC, StaticBAC V1, and StaticBAC V2

5. RESULTS

StaticBAC V2 Compression Gain (%) for 3 quantizers			
Model	**OLD**	**MSE**	**RD λ 0.15**
ViT-B/16	36.92	18.4	39.83
EfficientNetB7	21.87	21.96	42.5
ResNet50	46.7	24.72	41.78
GPT	17.3	31.86	45.23
BERT	17.47	17.53	39.28
SwinV2-B	15.33	17	39.15
VGG-19	7.19	7.19	38.42
MobileNet-V3	16.91	17	40.12
Inception-V3	22.42	22.44	41.07
T5	18.89	18.89	39.04
roBERTa	14.15	15.31	39.03
GPT-2	41.71	19.62	39.48
Llama 3.2 - 1B	15.67	15.67	39.11

Fig. 2. Average compression per model under three quantizers. Important: the gains for RD consider a fixed λ=0.15 for experimental consistency. This λ yields vastly different reconstruction errors per model, therefore the choice of the best λ should be taken individually aiming the least inference loss. We consider 0.15 to be the upper limit for this quantizer, thus 40% is the max compression and the MSE column represents the min compression.

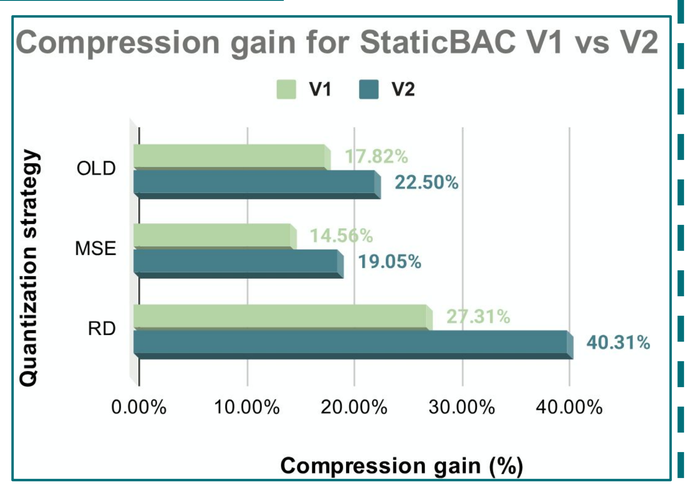


Fig. 3. Average compression gains across 13 NNs models and 3 quantizers for StaticBAC V1 and V2.

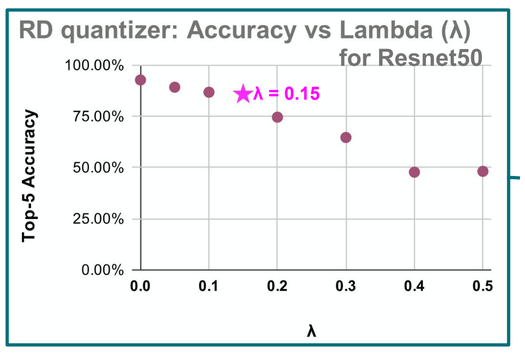


Fig. 4. Lambda sweep. With 85% accuracy, λ=0.15 represents a favorable rate-accuracy tradeoff. (for Resnet50)

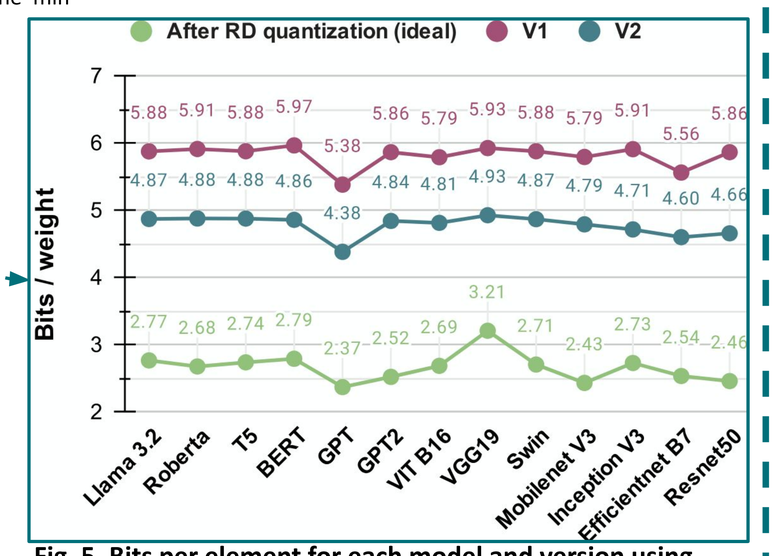


Fig. 5. Bits per element for each model and version using λ=0.15. Ideal means the Shannon entropy after quantization.

6. CONCLUSION

- **Up to 13% additional gain** for StaticBAC V2 across all quantization schemes.
  - V2's algorithmic modifications enable more compression and **stable coding performance** across diverse NN architectures with RD quantization.
- **40% average compression gain** using RD quantization: **better parameter distributions** for the entropy coder.
- V2 COST: **slower encoding**. Distribution dependent.
  - MSE quant.: encoder 66% slower and decoder 40% slower
  - RD quant.: encoder 21.4% slower and decoder 10% faster
- KEY takeway: **the quantizer and entropy coder should be designed together for the best compression result.**

REFERENCES:

[1] J. S. Gomes, M. Verhelst, F. L. L. Ramos and S. Bampi, "**StaticBAC: Fast and Memory-Efficient Entropy Coding for Neural Network Compression**," presented at the 39th SBC/SBMicro/IEEE Symposium on Integrated Circuits and Systems Design (SBCCI), São Paulo, Brazil, Aug. 2026.

[2] S. Wiedemann et al. **DeepCABAC: A universal compression algorithm for deep neural networks**. IEEE Journal on Selected Topics in Signal Processing, v. 17, n. 1, p. 212–225, 2023.

Code available: